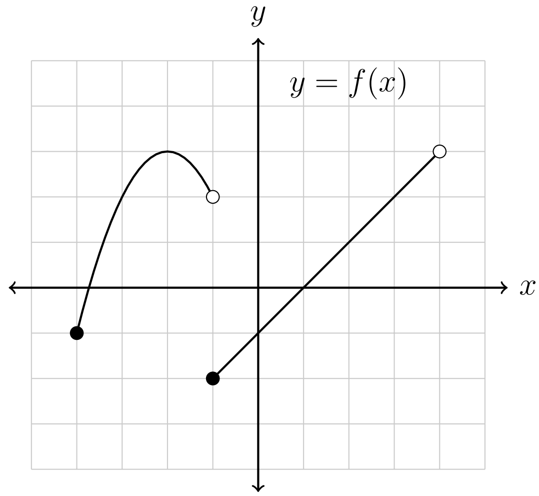
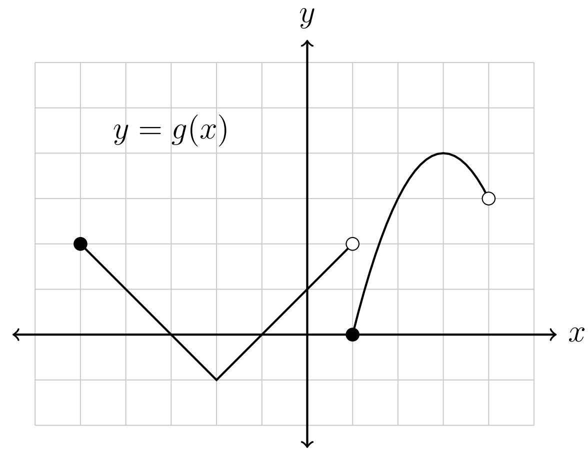
[5] **1.** Let $f(x)$ and $g(x)$ be given by the graphs below:





(a) State the domain of f in interval notation.

(b) State the range of f in interval notation.

(c) Evaluate: $(f \circ g)(2)$

(d) Evaluate: $g^{-1}(4)$

(e) Does f have an inverse function? Why or why not?

[4] **2.** Use polynomial long division to divide: $(2x^4 + 4x^3 - 5x^2 - 7x - 1) \div (x^2 - 3)$.

[3] **3.** Factor completely: $2x^4 - 5x^3 + 16x - 40$

[3] **4.** Complete the square: $x^2 + 8x + 5$

[4] **5.** Given the function $f(x) = -x^2 + 4x + 5$,

(a) Find the coordinates of all axis intercepts.

(b) Find the coordinates of the vertex.

(c) Sketch a graph on the given axes and label the points found in the previous parts.

[4] **6.** Given $f(x) = \dfrac{2(x-7)(x+5)}{x(x+3)(x-5)}$, $g(x) = \dfrac{(2x+1)(x-7)}{(3x-2)(x-3)}$, and $h(x) = \dfrac{x(x+3)(x+7)}{(x+8)(x+5)}$, state the domain and simplify the following:

(a) $(f \cdot h)(x)$

(b) $(f/g)(x)$

[4] **7.** Given $f(x) = \dfrac{4}{2x+5}$

and $g(x) = \dfrac{x+7}{2x^2+13x+20}$

(a) Simplify: $(f+g)(x)$.

(b) State the domain of $(f+g)(x)$.

8. Given $f(x) = \dfrac{3x-5}{8x-3}$ and $g(x) = \dfrac{1}{x+1}$

[3] (a) Simplify $(f \circ g)(x)$.

[3] (b) Find $f^{-1}(x)$.

[2] **9.** State the domain of the function in interval notation.

$f(x) = \dfrac{\sqrt{8-x}}{\sqrt{x-2}}$

10. Solve the following equations for x:

[4] (a) $\dfrac{x+4}{x+1} - 5 = \dfrac{2x}{x+1}$

[4] (b) $\sqrt{x+1} - \sqrt{x^2-5} = 0$

[3] **11.** Simplify and write your answer without negative exponents (assume that $x, y, z > 0$) : $\dfrac{(\sqrt[3]{x^6y^{12}}\sqrt{xz^{-10}})^5}{x^4\sqrt{x^2+9}}$

[3] **12.** Solve the inequality and express your answer in interval notation.

$\dfrac{(x-7)(x+3)^2}{5x} \geqslant 0$

[3] **13.** How much money should you invest today into an account which pays 6% interest compounded monthly in order to have \$15000 in 4 years? (Round your answer to two decimal places.)

[4] **14.** Given $f(x) = \left(\frac{1}{3}\right)^x + 9$

(a) State the equation of any asymptote of the function f.

(b) Find the coordinates of all axis intercepts of f.

(c) Sketch $y = f(x)$ on the axes provided. Clearly display the information you found in the previous parts.

[2] **15.** Calculate $\log_5(13)$ to two decimal places.

[3] **16.** Express as a single logarithm: $\dfrac{1}{2}\ln x + 2\ln(x+1) - 3\ln(y-1)$

17. Solve for x:

[4] (a) $64(4^x) = 8^{2x-1}$

[4] (b) $\log_2(x) + \log_2(x+3) = 2$

[4] (c) $2^{x+2} = 3^{x+3} \cdot 5$. (Give an answer in the form $\frac{\ln A}{\ln B}$)

[3] **18.** Given $\cos(\theta) = \frac{3}{5}$;

(a) Find $\tan(\theta)$ if θ is acute.

(b) Find $\tan(\theta)$ if θ is in the fourth quadrant.

[2] **19.** Convert $\frac{7\pi}{12}$ radians to degrees.

20. For the angle $\theta = \frac{7\pi}{6}$;

[2] (a) Sketch θ in standard position on the given axes.

[2] (b) Find the exact value of $\csc\left(\frac{7\pi}{6}\right)$.

[3] **21.** Find all θ in $[0, 2\pi)$ such that $\cot(\theta) = \sqrt{3}$.

[4] **22.** Prove the identity $\dfrac{\csc^2 x - \sin^2 x}{\csc x + \sin x} = \cos x \cot x$

[4] **23.** State the amplitude, period, and sketch at least two cycles of the function $f(x) = -3\cos(4x)$.

[4] **24.** A triangle has sides of length a, b and c across from angles of measure A, B, and C respectively. Suppose $b = 8$, $c = 6$, and $A = 30°$. Find a, B and C. Give your answers rounded to one decimal place.

[3] **25.** If you stand 840 feet away from the Empire State Building, the angle of elevation to the top of the building is 60°. How tall is the Empire State Building? (Answer in feet rounded to one decimal place.)

ANSWERS:

1(a) $[-4, 4)$

1(b) $[-2, 3]$

1(c) $(f \circ g)(2) = 2$.

1(d) $g^{-1}(4) = 3$.

1(e) No. It fails the horizontal line test.

2. $2x^2 + 4x + 1 + \dfrac{5x+2}{x^2-3}$

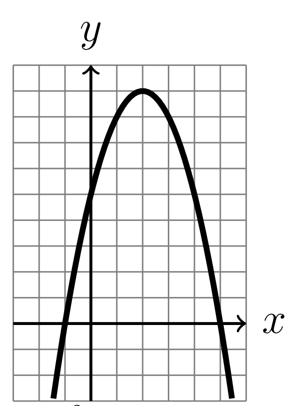
3. $(x+2)(x^2-2x+4)(2x-5)$

4(a) y-int: $(0, 5)$,

x-int's: $(-1, 0), (5, 0)$

4(b) vertex: $(2, 9)$

4(c)



5. $(x+4)^2 - 11$

6(a) $\frac{2(x-7)(x+7)}{(x-5)(x+8)}, \mathbb{R}\backslash\{-8,-5,-3,0,5\}$

6(b) $\frac{2(x+5)(3x-2)(x-3)}{x(x+3)(x-5)(2x+1)}$,

$\mathbb{R}\backslash\{-3,-\frac{1}{2},0,\frac{2}{3},3,5,7\}$

7(a) $\frac{5x+23}{(2x+5)(x+4)}$

7(b) $\mathbb{R}\backslash\{-4,\frac{-5}{2}\}$

8(a) $\frac{-2-5x}{5-3x}$

8(b) $\frac{3x-5}{8x-3}$

9. $\dfrac{x^{17/2}y^{20}}{z^{25}\sqrt{x^2+9}}$

10. $(2,8]$

11(a) $x=\frac{-1}{6}$

11(b) $x=3$

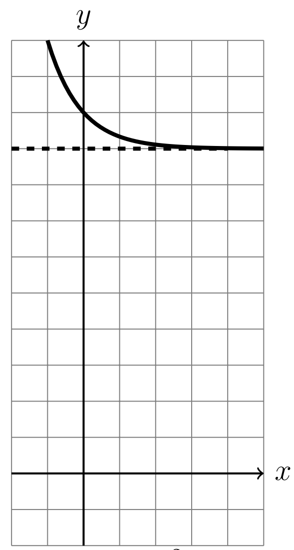
12. $(-\infty,0)\cup[7,\infty)$

13. \$11806.48

14(a) H.A. at $y=9$

14(b) y-int: $(0,10)$, no x-int

14(c)



15. $\ln\left(\dfrac{\sqrt{x}(x+1)^2}{(y-1)^3}\right)$

16. 1.59

17(a) $x=\dfrac{9}{4}$

17(b) $x=1$

17(c) $x=\frac{\ln\left(\frac{135}{4}\right)}{\ln\left(\frac{2}{3}\right)}$

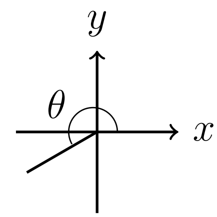
18(a) $\tan\theta=4/3$

18(b) $\tan\theta=-4/3$

19. $\frac{7\pi}{12}=105°$

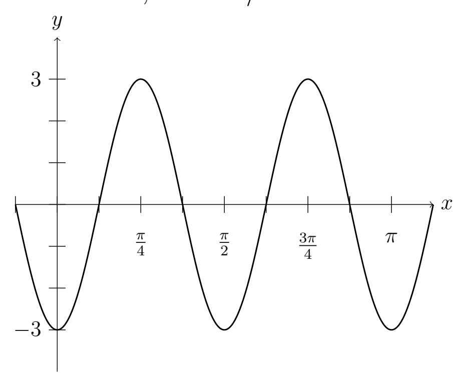
20. $\theta=\frac{\pi}{6},\theta=\frac{7\pi}{6}$

21(a)



21(b) $\csc\left(\frac{7\pi}{6}\right)=-2$

23. $A=3$, $P=\pi/2$.



24. $a\approx 4.1, C\approx 47.0°, B\approx 103.0°$

25. 1454.9ft.